

Module 3-Part 2

- Three-dimensional viewing pipeline. Projections- Parallel and Perspective projections. Visible surface detection algorithms- Depth buffer algorithm, Scan line algorithm

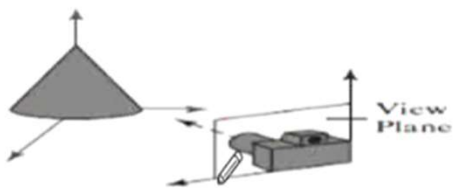
1

Overview of 3D Viewing

- Three-dimensional viewing involves some tasks that are not present in two-dimensional viewing.
- For example, projection routines are needed to transfer the scene to a view on a planar surface, visible parts of a scene must be identified
- For a realistic display, lighting effects and surface characteristics must be taken into account.
- To obtain a display of a three-dimensional world-coordinate scene, we first set up a coordinate reference for the viewing, or "camera," parameters.
- This coordinate reference defines the position and orientation for a view plane (or projection plane) that corresponds to a camera film plane.

2

Overview of 3D Viewing

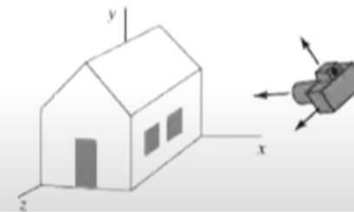


- Object descriptions are then transferred to the viewing reference coordinates and projected onto the view plane.
- We can generate a view of an object on the output device in wireframe (outline) form
- We can apply lighting and surface-rendering techniques to obtain a realistic shading of the visible surfaces.

3

3D Viewing Pipeline

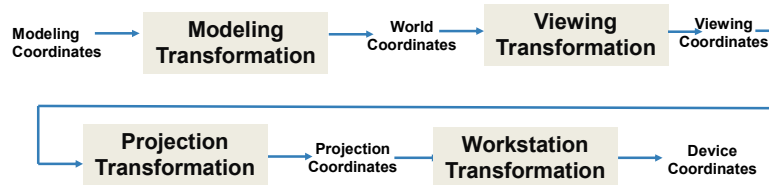
- We need to choose a viewing position corresponding to where we would place a camera.
- We choose the viewing position according to whether we want to display a front, back, side, top, or bottom view of the scene.
- Then we must decide on the camera orientation.



4

3 D Viewing Pipeline

The steps for computer generation of a view of a three dimensional scene are somewhat analogous to the processes involved in taking a photograph.



5

- Construct the shape of individual objects in a scene within modeling coordinate
- Deciding the dimensions of how much of the scene to capture is **modeling transformation** ,and place the objects into appropriate positions within the scene in order to fit in the frame(**world coordinate**)
- Setting the position and orientation of the camera is **viewing transformation** .
- Appropriate scaling up or down is done so that proper view become available in camera window coordinate (**viewing coordinate**).
- Convert the viewing coordinate description of the scene to coordinate positions on the projection plane by adjusting focusing of camera and direction of light is **projection transformation** , so that it forms an image in the camera frame or film (**projection coordinates**)
- Window to view port transformation, while clicking the camera button image is formed in camera screen (**device coordinate**).

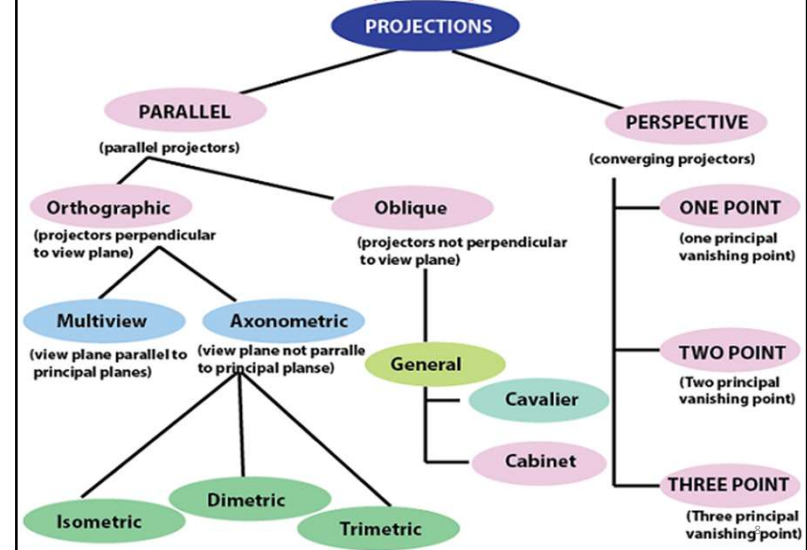
6

Projections

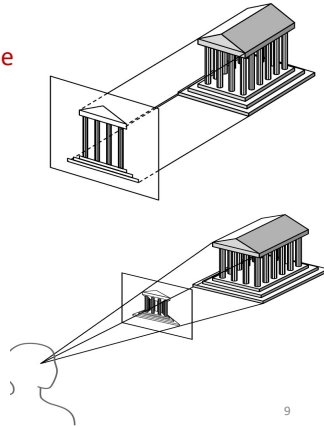
- Projection is mapping or transformation from 3D view to 2D view.
- 2 types of Projections
 - Parallel
 - Perspective
- Projected view of an object is determined by calculating the intersection of the projection lines with the view plane(displayed surface).

7

Taxonomy of Projections



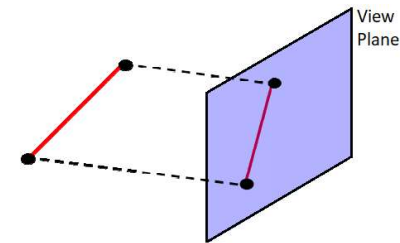
- **Projections** transform points in n -space to m -space, where $m < n$.
- In 3-D, map points from 3D-space to the **projection plane** (PP) (image plane) along **projectors** (viewing rays) emanating from the center of projection (COP)
- Standard projections project **onto a plane**
- **Projectors** are lines that either
 - converge at a center of projection
 - are parallel



9

Parallel Projections

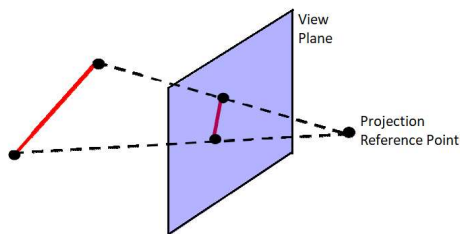
- In Parallel projection, coordinate positions are transformed to the view plane along parallel lines



10

Perspective Projections

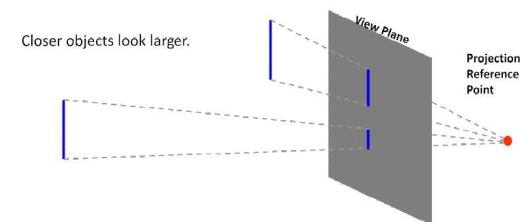
- In perspective projection, object position are transformed to the view plane along lines that converge to a point called projection reference point (center of projection)



11

Perspective Projections

- Produces realistic views but does not preserve relative proportions
- Projection of distant objects are smaller than the projection of objects of the same size that are closer to the projection plane



12

Parallel Projection

- It preserves relative proportion of object.
- Accurate views of the various sides of an object are obtained with a parallel projection.
- Does not give a realistic representation of the appearance of a three-dimensional object.

13

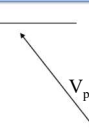
Parallel Projections

- We can define a parallel projection with a projection vector, V_p that defines the direction for the projection lines.
- 2 types:
 - **Orthographic** : when the projection is perpendicular to the view plane.
 - **Oblique** : when the projection is not perpendicular to the view plane.

Orthographic projection



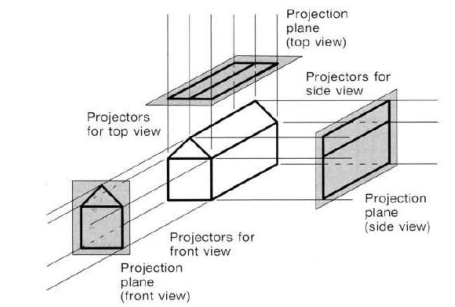
Oblique projection



14

Orthographic projections

- Orthographic projection produce front, side and rear views of an object



15

Orthographic projections

- Front, side and rear orthographic projection of an object are called **elevations** and the top orthographic projection is called **plan view**.

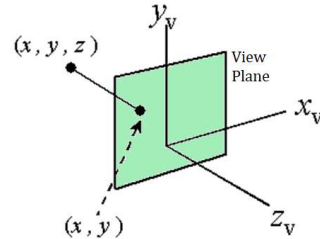
16

Transformation equations for an Orthographic Projection

If the view plane is placed at position z_{vp} along the z_v axis, then any point (x, y, z) in viewing coordinates is transformed to projection coordinates as

$$x_p = x, \quad y_p = y$$

where the original z -coordinate value is preserved for the depth information needed in depth cueing and visible surface determination procedures.

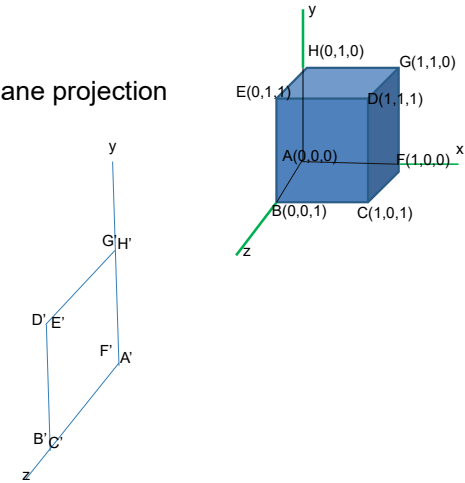


17

Find the orthographic projection of a unit cube onto the $x=0, y=0$ and $z=0$ plane.

X=0 plane projection /YZ plane projection

$A(0,0,0) - A'(0,0,0)$
 $B(0,0,1) - B'(0,0,1)$
 $C(1,0,1) - C'(0,0,1)$
 $D(1,1,1) - D'(0,1,1)$
 $E(0,1,1) - E'(0,1,1)$
 $F(1,0,0) - F'(0,0,0)$
 $G(1,1,0) - G'(0,1,0)$
 $H(0,1,0) - H'(0,1,0)$

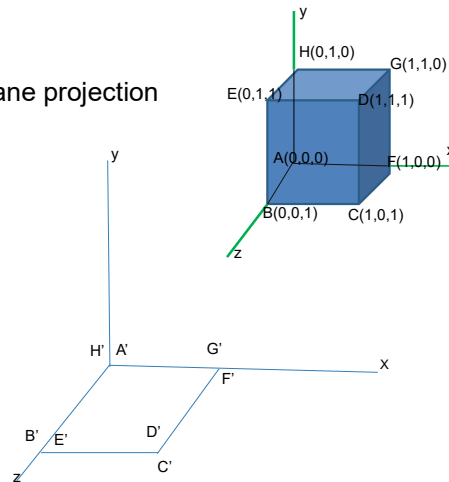


18

Find the orthographic projection of a unit cube onto the $x=0, y=0$ and $z=0$ plane.

Y=0 plane projection /XZ plane projection

$A(0,0,0) - A'(0,0,0)$
 $B(0,0,1) - B'(0,0,1)$
 $C(1,0,1) - C'(1,0,1)$
 $D(1,1,1) - D'(1,0,1)$
 $E(0,1,1) - E'(0,0,1)$
 $F(1,0,0) - F'(1,0,0)$
 $G(1,1,0) - G'(1,0,0)$
 $H(0,1,0) - H'(0,0,0)$

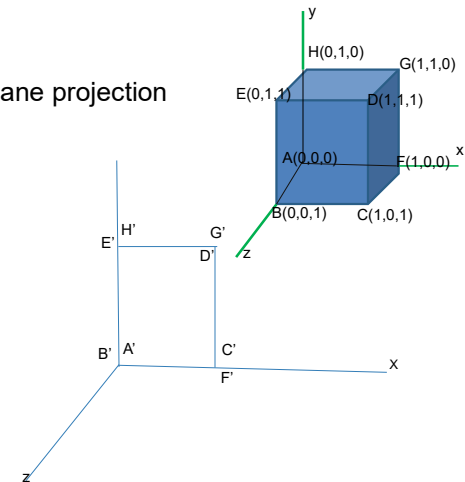


19

Find the orthographic projection of a unit cube onto the $x=0, y=0$ and $z=0$ plane.

Z=0 plane projection /XY plane projection

$A(0,0,0) - A'(0,0,0)$
 $B(0,0,1) - B'(0,0,0)$
 $C(1,0,1) - C'(1,0,0)$
 $D(1,1,1) - D'(1,1,0)$
 $E(0,1,1) - E'(0,1,0)$
 $F(1,0,0) - F'(1,0,0)$
 $G(1,1,0) - G'(1,1,0)$
 $H(0,1,0) - H'(0,1,0)$



20

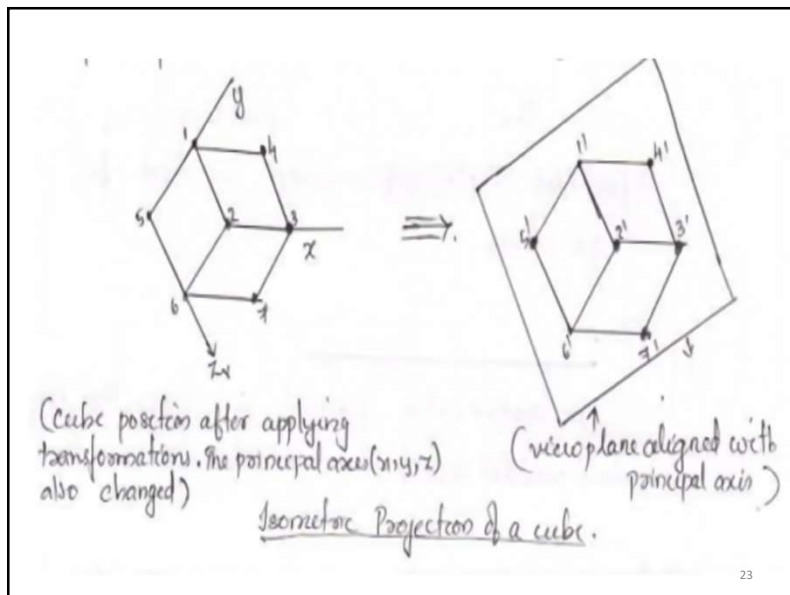
Axonometric orthographic projections.

- Orthographic projections that *show more than one face of an object i.e; atleast 3 faces are shown by manipulating the object (applying basic transformation on the object) are called **Axonometric orthographic projections.***
- Three varieties:
 - **Isometric**
 - **Dimetric**
 - **Trimetric**

21

- Foreshortening factors- It is the ratio of projected length to true length of a line
- **Trimetric**- All the three foreshortening factors are different
- **Dimetric**- Two foreshortening factors are equal
- **Isometric**- All the three foreshortening factors are equal
 - The most common axonometric projection is an **isometric projection** where the projection plane intersects each coordinate axis in the model coordinate system at an equal distance.
 - In this projection parallelism of lines are preserved but angles are not preserved.

22



23

Parallel Projection - Advantages and Disadvantages

Are actually a kind of **affine transformation**

- Preserves **distances**
- **Shapes preserved**
- Parallel lines remain **parallel**
- **Ratios** are preserved
- Angles not (in general) preserved
- **Not realistic looking**
 - because many **surfaces** are **hidden** from view
- **Good for exact measurements**
 - **Building plans** ,**Manuals**, **CAD**, **architectural drawings**

24

Parallel Projection:

- Coordinate positions are transformed to the view plane along parallel lines
- Preserves relative proportions of objects
- Used in drafting to produce scale drawings of 3D objects
- Accurate view of various sides of objects obtained
- Does not give a realistic representation of 3D object

Perspective Projection

- Object positions are transformed to the view plane along lines that converge to a point called the projection reference point (or center of projection).

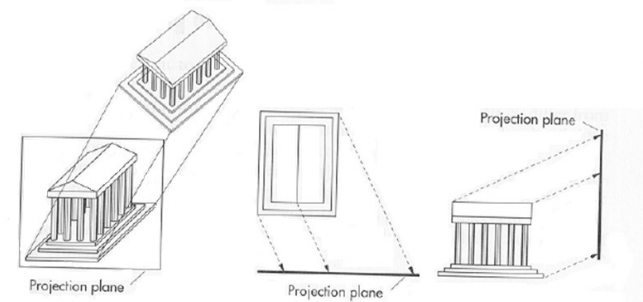
Does not preserve relative proportions

- The projected view of an object is determined by calculating the intersection of the projection lines with the view plane.
- Produce realistic views

25

Oblique Projections

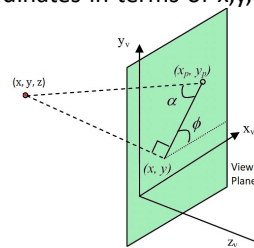
- Oblique projection is obtained by projecting points along parallel lines that are not perpendicular to the projection plane.



26

Transformation equations for oblique projection

- Point (x, y, z) is projected to position (x_p, y_p) on the view plane xy .
- *Orthogonal projection coordinates on the plane are (x, y)*
- The oblique projection line from (x, y, z) to (x_p, y_p) makes an angle α with the line on the projection plane that joins (x_p, y_p) and (x, y) .
- This line, of length L , is at an angle ϕ with the horizontal direction in the projection plane.
- We can express the projection coordinates in terms of x, y, L and ϕ as:



27

Transformation equations for oblique projection

$$\cos \phi = \frac{x_p - x}{L}$$

$$\sin \phi = \frac{y_p - y}{L}$$

$$x_p = x + L \cos \phi \quad \text{-----(1)}$$

$$y_p = y + L \sin \phi \quad \text{-----(2)}$$

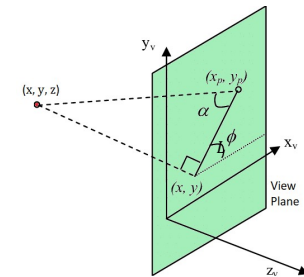
- Length L depends on the angle α and the z coordinate of the point to be projected

$$\tan \alpha = \frac{z}{L}$$

$$L = \frac{z}{\tan \alpha}$$

$$= zL_1$$

$$\text{where } L_1 = \frac{1}{\tan \alpha} \quad (L=L_1 \text{ when } \alpha=1)$$



28

Transformation equations for oblique projection

Rewrite the equations (1) and (2) as

$$x_p = x + z(L_1 \cos \phi) \quad \text{-----(3)}$$

$$y_p = y + z(L_1 \sin \phi) \quad \text{-----(4)}$$

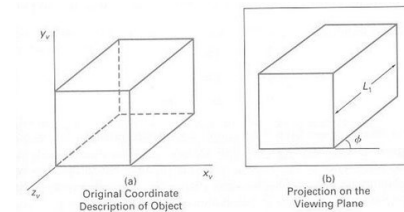
The transformation matrix can be written as:

$$\begin{bmatrix} 1 & 0 & L_1 \cos \phi & 0 \\ 0 & 1 & L_1 \sin \phi & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

29

Oblique projections

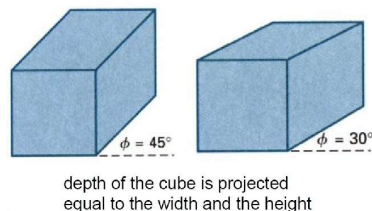
- There are two kinds of oblique projections:
 - Cavalier projection
 - Cabinet projection



30

Cavalier Projection

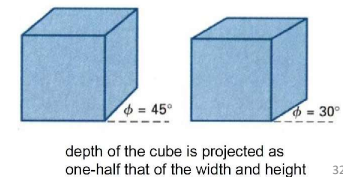
- All lines perpendicular to the projection plane are projected with no change in length.
- Projection line makes an angle 45 degree with projection plane $\alpha = 45^\circ$
- Common choice for ϕ is usually 30° or 45°



31

Cabinet Projection

- Lines which are perpendicular to the projection plane (viewing surface) are projected at $\frac{1}{2}$ the length
- This results in foreshortening of the z axis, and provides a more “realistic” view than cavalier projections.
- Projection line makes an angle 63.4 degree with projection plane, $\alpha = 63.4$
- Cabinet projections appear more realistic than cavalier projections because of reduction in the length of perpendiculars



32

Transformation equations Perspective Projection

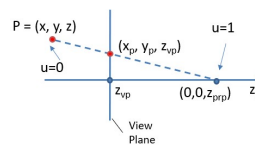
- To obtain a perspective projection of a three dimensional object, we transform points along projection lines that meet at the projection reference point.
- Suppose we set the projection reference point (PRP) at z_{prp} along the z_v axis, and we place the view plane at z_{vp} as in figure.
- We can write the equations describing coordinate positions along this perspective projection line in parametric form as

$$x' = x - xu \text{ -----(1)}$$

$$y' = y - yu \text{ -----(2)}$$

$$z' = z - (z - z_{prp})u \text{ -----(3)}$$

- Parameter u takes value from 0 to 1 and coordinate position (x', y', z') represents any point along the projection line



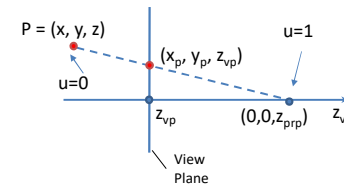
Transformation equations Perspective Projection

- When $u=0$, we are at position $P=(x, y, z)$. At the other end of the line, $u=1$ and we have the projection reference point coordinates $(0, 0, z_{prp})$.
- On the view plane

$$z' = z_{vp} \text{ -----(4)}$$

substitute (4) in (3) and rearranging

$$\text{then } u = \frac{z_{vp} - z}{z_{prp} - z}$$



34

- Substituting u for a point on view plane

$$x_p = x - xu = x(1 - u) = x \left(1 - \frac{z_{vp} - z}{z_{prp} - z} \right)$$

$$= x \left(\frac{z_{prp} - z - z_{vp} + z}{z_{prp} - z} \right)$$

$$= x \left(\frac{z_{prp} - z_{vp}}{z_{prp} - z} \right) = x \left(\frac{d_p}{z_{prp} - z} \right)$$

$$y_p = y - yu = y(1 - u) = y \left(1 - \frac{z_{vp} - z}{z_{prp} - z} \right)$$

$$= y \left(\frac{z_{prp} - z - z_{vp} + z}{z_{prp} - z} \right)$$

$$= y \left(\frac{z_{prp} - z_{vp}}{z_{prp} - z} \right) = y \left(\frac{d_p}{z_{prp} - z} \right)$$

where $d_p = z_{prp} - z_{vp}$ is the distance of the view plane from the projection reference point.

35

- Using a three-dimensional homogeneous coordinate representation, we can write the perspective projection transformation in matrix as

$$\begin{bmatrix} x_h \\ y_h \\ z_h \\ h \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -z_{vp}/d_p & z_{vp}(z_{prp}/d_p) \\ 0 & 0 & -1/d_p & z_{prp}/d_p \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

36

- If the projection reference point is fixed at the coordinate origin and $(x_{prp}, y_{prp}, z_{prp}) = (0, 0, 0)$

$$x_p = x \left(\frac{z_{prp} - Z_{vp}}{z_{prp} - z} \right) = x \left(\frac{Z_{vp}}{z} \right)$$

$$y_p = y \left(\frac{z_{prp} - Z_{vp}}{z_{prp} - z} \right) = y \left(\frac{Z_{vp}}{z} \right)$$

37

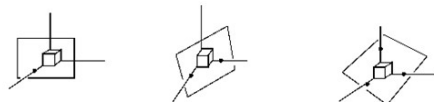
Vanishing Point

- Parallel lines that are parallel to the view plane will be projected as parallel lines.
- Any set of parallel lines of objects that are not parallel to the projection plane are projected into converging lines.
- The point at which set of projected parallel lines appears to converge is called vanishing point.
- A different set of projected parallel lines will have a separate vanishing point.
- A scene can have any number of vanishing points depending on sets of parallel lines in a scene.

38

Vanishing Point

- Vanishing point for any set of lines that are parallel to one of the principal axes of an object is referred to as the **principal vanishing point**.
- There are at most 3 such points, corresponding to the number of axes cut by the projection plane.
- The number of principal vanishing points is determined by the number of principal axes intersected by the view plane.



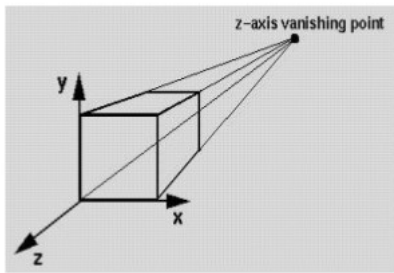
39

Classes of Perspective Projection

- We can control the number of principal vanishing points with the orientation of the projection plane & perspective projection are accordingly classified.
 - **One-Point Perspective Projection**
 - **Two-Point Perspective Projection**
 - **Three-Point Perspective Projection**
- Appearance of the object changes depending on these projections.

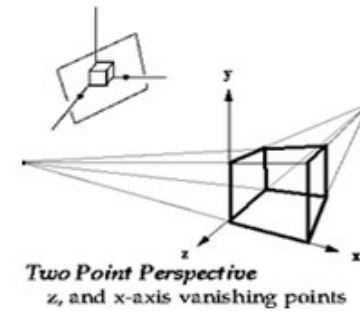
40

One-Point Perspective



41

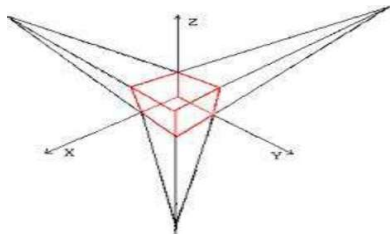
Two-Point Perspective



This is often used in architectural, engineering and industrial design drawings.

42

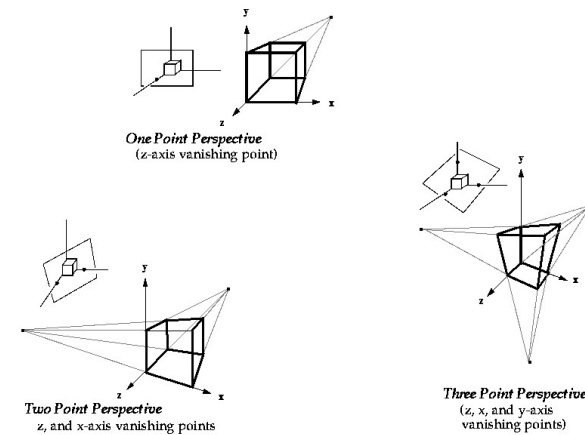
Three-Point Perspective



Three-point perspective projection is used less frequently as it adds little extra realism to that offered by two-point perspective projection

43

Vanishing Point



44

Visible Surface Detection Algorithms (Hidden surface removal/ hidden surface detection)

- A major consideration in the generation of realistic graphics displays is identifying those parts of a scene that are visible from a chosen viewing position.
- Various algorithms have been devised for efficient identification of visible objects for different types of applications.
- Some methods require more memory, some involve more processing time, and some apply only to special types of objects.
- Deciding upon a method for a particular application can depend on such factors as the complexity of the scene, type of objects to be displayed, available equipment, and whether static or animated displays are to be generated.

45

Visible Surface Detection Algorithms

- The various algorithms are referred to as **visible-surface detection methods**.
- Sometimes these methods are also referred to as **hidden-surface elimination methods**.
- Visible-surface detection algorithms are broadly **classified** according to whether they deal with **object definitions directly or with their projected images**.
- These two approaches are called **object-space methods** and **image-space methods**, respectively.

46

Visible Surface Detection Algorithms

An object-space method:

- compares objects to each other within the scene definition to determine which surfaces we should label as visible
- Used in vector graphics systems
- Used for accuracy
- Continuous operation
- E.g.:- Back face method

An Image-space algorithm

- visibility is decided point by point at each pixel position on the projection plane.
- Used in raster scan systems
- Less processing time
- Discrete operation
- E.g.:- Depth buffer method

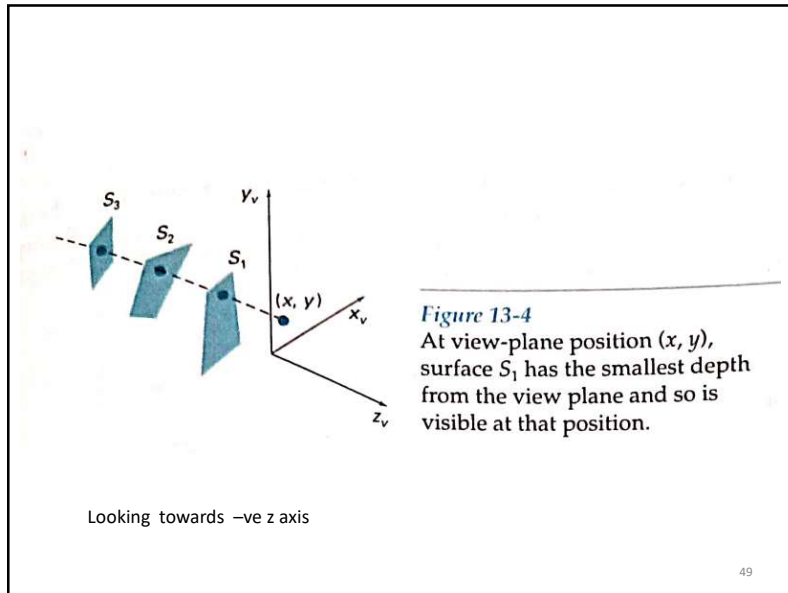
Most visible-surface algorithms use image-space methods.

47

Depth –buffer(z-buffer) method

- A commonly used image-space approach to detecting visible surfaces is the **depth-buffer method**, which compares surface depths at each pixel position on the projection plane.
- Due to depth calculation it is referred as depth buffer.
- This procedure is also referred to as the z-buffer method, since object depth is usually measured from the view plane along the z axis of a viewing system.

48



- Each surface of a scene is processed separately, one point at a time across the surface.
- The method is usually applied to scenes containing only polygon surfaces, because depth values can be computed very quickly and the method is easy to implement.
- Each (x, y, z) position on a polygon surface corresponds to the orthographic projection point (x, y) on the view plane.
- For each pixel position (x, y) on the view plane, object depths can be compared by comparing z values. then surfaces with highest depth(z value) is closer to the viewing position , so it is visible and its surface intensity value at (x, y) is saved for display.

- Two buffers are required. A depth buffer is used to store depth values for (x, y) position as surfaces are processed, and the refresh buffer stores the intensity values for each position.
- Initially all positions in the depth buffer are set to 0 (minimum depth), and the refresh buffer is initialized to the background intensity.
- Each surface listed in the tables is then processed, one scan line at a time, calculating the depth (z value) at each (x, y) pixel position. The calculated depth is compared to the value previously stored in the depth buffer at that position.
- If the calculated depth is greater than the value stored in the depth buffer, the new depth value is stored, the surface intensity at that position is determined and placed in the same xy in the refresh buffer.

Depth Buffer Algorithm

1. Initialize the depth buffer and refresh buffer so that for all buffer positions (x, y) ,
 $\text{depth}(x, y) = 0, \text{ refresh}(x, y) = I_{\text{background}}$
 2. For each position on each polygon surface, compare depth values to previously stored values in the depth buffer to determine visibility.
 - Calculate the depth z for each (x, y) position on the polygon.
 - If $z > \text{depth}(x, y)$, then set
 $\text{depth}(x, y) = z, \text{ refresh}(x, y) = I_{\text{surf}}(x, y)$
- where,
- $I_{\text{background}}$ = background intensity
 $I_{\text{surf}}(x, y)$ = projected intensity value for the surface at pixel position (x, y) .
- After all surfaces have been processed, the depth buffer contains depth values for the visible surfaces and the refresh buffer contains the corresponding intensity values for those surfaces.

Equation for a plane is

$$Ax + By + Cz + D = 0$$

- Depth values for a surface position(x, y) are calculated from the plane equation for each surface by:

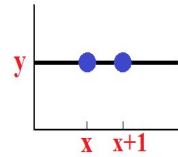
$$z = \frac{-Ax - By - D}{C} \quad \text{-----(1)}$$

- If the depth of the position (x,y), has been determined to be z then the depth z' of the next position (x+1,y) along the scan line is obtained as

$$\begin{aligned} z' &= \frac{-A(x+1) - By - D}{C} \\ &= \frac{-Ax - A - By - D}{C} \quad \text{-----(2)} \end{aligned}$$

substitute 1 in 2

$$z' = z - \frac{A}{C} \quad \text{-----(3)}$$



53

- The ratio $-A/C$ is constant for each surface, so succeeding depth values across a scan line are obtained from preceding values with a single addition.
- On each scan line, we start by calculating the depth on a left edge of the polygon that intersects that scan line. Depth values at each successive position across the scan line are then calculated by equation 3

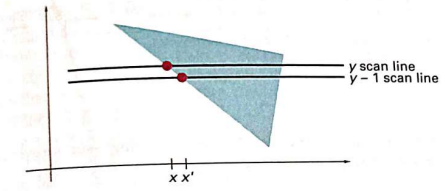


Figure 13-7
Intersection positions on successive scan lines along a left polygon edge.

4

- Vertically next polygon depth point is at $(x - \frac{1}{m}, y-1)$
- We can find value depth value z' at this point by substituting $(x - \frac{1}{m}, y-1)$ in equation (1)

$$\text{ie, } z' = \frac{-Ax - By - D}{C}$$

$$= \frac{-A(x - \frac{1}{m}) - B(y-1) - D}{C} = \frac{-Ax + \frac{A}{m} - By + B - D}{C}$$

$$= \frac{-Ax - By - D}{C} + \frac{\frac{A}{m} + B}{C}$$

$$z' = z + \frac{A/m + B}{C} \quad \text{-----(4)}$$

If we are processing down a vertical edge, slope is infinite and (4) reduces to

$$z' = z + \frac{B}{C} \quad \text{-----(5)}$$

55

Advantages:

- Simple to use
- Can be implemented easily in object or image space

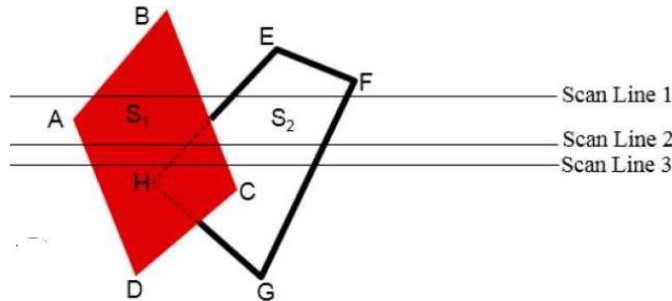
Disadvantages:

- Takes up a lot of memory
- It is a time-consuming process

56

Scan Line Method

- It is an image space method to identify visible surfaces.
- This is an extension of the scan line algorithm for the polygon filling algorithm.



57

Scan Line Method

- We can process only one surface at a time if we use depth buffer method.
- But scan line method can process **more than one surfaces at a time**.
- This method has depth information for the only single scan line.
- In order to require one scan-line of depth values, we must group and process all polygons intersecting a given scan-line at the same time before processing the next scan-line
- Two important tables, **edge table and polygon table**, are maintained for this

58

Scan Line Method

- **The Edge Table** – It contains
 - coordinate endpoints of each line in the scene,
 - the inverse slope of each line,
 - pointers into the polygon table to connect edges to surfaces.
- **The Polygon Table** – It contains
 - coefficients of the plane equation for each surface,
 - Intensity information for the surfaces,
 - a flag that is initialized to false

59

Scan Line Algorithm:

- For all polygons (surfaces) intersecting a scan line, the faces are processed from left to right.
- If there are overlapping faces, we should determine the depth so that nearest face to the view plane is identified and the intensities of that face is entered into the refresh buffer.
- For each scan line an active list of edges {edges intersecting by scan line} of faces is maintained and they are sorted in the order of **increasing x values**.

60

- **Active list of edges**

- Stores those edges that cross the current scan-line
 - Sorted in order of increasing x.

61

Flag for each surface

- Indicate whether inside or outside of the surface
 - At the leftmost boundary of a surface
 - The surface flag is turned on
 - At the rightmost boundary of a surface
 - The surface flag is turned off

62

- **Active list for scan line 1**

- Edge table

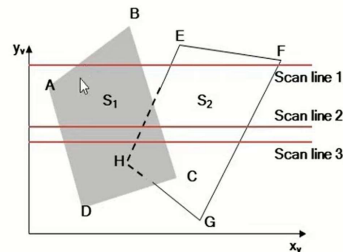
- AB, BC, EH, and FG

- Between AB and BC, only

the flag for surface S_1 is on

- No depth calculations are necessary
- Intensity for surface S_1 is entered into the refresh buffer

- Similarly, between EH and FG, only the flag for S_2 is on



63

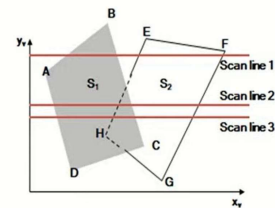
- **For scan line 2**

- AD, EH, BC, and FG

- Between AD and EH, only the flag for S_1 is on
- Between EH and BC, the flags for both surfaces are on
 - Depth calculation is needed using plane coefficient.
 - If depth $S_1 < S_2$ Intensities for S_1 are loaded into the refresh buffer until BC
 - After that S_1 intensity of surface S_2 is stored until FG

- Scan line 3 has the same active list as scan line 2

- Unnecessary to make depth calculations between EH and BC



64

Advantages

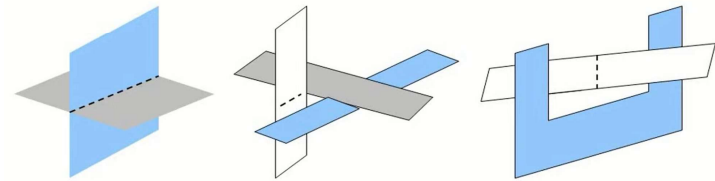
- Any number of overlapping surfaces can be processed , Flags for the surfaces are set to indicate whether a position is inside or outside, and depth calculations are performed when surfaces overlap
- Take advantage of coherence , When these coherence methods are used, we need to be careful to keep track of which surface section is visible on each scan line.
- This works only if surfaces do not cut through or otherwise cyclically overlap each other

disadvantages

- It does not work for the surface that cut through or cyclic overlap each other

65

- If any kind of cyclic overlap is present in a scene, we can divide the surfaces to eliminate the overlaps.
- The dashed lines in this figure indicate where planes could be sub divided to form two distinct surfaces, so that the cyclic overlaps are eliminated.



66